BU433 Case 04 Submission

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> **Note**
> I wasn't able to attend the class when the case was discussed. I have however read the case, and attempted to make my answers as detailed as possible.

Q1.

- **Asset-liability mismatch:** During the tech boom of 2020–2021, SVB's deposits surged from $62 billion to $189 billion.
- **The Fed's Rate Hikes:** When the Federal Reserve rapidly hiked interest rates to combat inflation, the market value of SVB's long-term bonds fell massively.
- **Sector Concentration:** SVB's depositors were heavily concentrated in venture capital-backed startups. When venture funding dried up in late 2022, these startups began steadily withdrawing their deposits to fund operations.
- **Twitter-Fueled Bank Run:** Worried about losses, prominent VCs instructed their portfolio companies to pull their funds. Depositors withdrew $42 billion in a single day using online banking. This forced regulators to step in and shut down the insolvent bank.
- **Risk Management Failures:** SVB operated without a Chief Risk Officer for nine months and repeatedly ignored its own internal interest-rate risk limits and warnings.

Q2.

- **Implications of Focussing on the Tech Industry**
  - **Extreme Covariance:** SVB's liabilities were completely tied to the macroeconomic health of the tech sector. When tech boomed in 2021, deposits increased exponentially. However, in 2022, when VC funding dried up, the entire client base began burning cash and withdrawing capital simultaneously.
  - **Network Effects:** The VC community acted like an echo chamber. This allowed a handful of VC heads to instruct their entire portfolios to pull funds

simultaneously via coordinated communication on social media platforms like Twitter.

- **Managing and Avoiding the Risk**
  - **Implement Asset-Liability Management:** SVB should have kept its asset duration short and highly liquid to match its volatile and short-term deposit liabilities.
  - **Aggressive Interest Rate Hedging:** If SVB's management insisted on holding long-term fixed assets, they should have participated in interest rate swaps to hedge against duration risk.
  - **Deposit Diversification:** SVB could have diversified their client base to non-tech business lines to build a more uncorrelated portfolio. Even in the tech sector, they could have limited the exposure to single VC ecosystems to minimize damage due network effect.

Q3.

- **What has changed over the past few years?**
  - Total deposits first grew from $49,992M in 2018 to $191,431M in 2021. Then fell to $175,378M in 2022. *(Exhibit 1)*
  - Total assets nearly quadrupled from $55.2B to $211.8B *(Exhibit 5)*
  - Fed funds rate grew from 0.1% in 2020 to 4.9% in 2023. It peaked at 9.1%. *(Exhibit 9)*
  - Long term bond yield rates rose sharply from 0.6% in 2020 to 4% in 2023. This directly eroded the value of SVB's existing fixed-rate bond holdings. *(Exhibit 12)*
  - ROA fell from 1.9% in 2018 to 0.8% in 2022. ROE fell from 21.1% to 9.8%. *(Exhibit 5)*
- **How did SVB respond to those changes?**
  - SVB bought most fixed-rate / long-duration securities before the March 2022 rate hikes. *(Exhibit 12)*
  - Hold-To-Maturity (HTM) securities grew from $16,592M in 2020 to $98,195M in 2021. *(Exhibit 11)*
  - SVB sold $21B of Available-For-Sale (AFS) securities at a $1.8B after-tax loss and attempted a $2.25B capital raise. *(Exhibit 15)*
- **What key risk management issues were faced by SVB?**
  - No Chief Risk Officer (CRO) from Apr 2022 to Jan 2023.
  - 31 open supervisory findings at the time of failure.
  - Repeatedly failed internal stress tests.
  - Category IV regulatory requirements hadn't been phased despite crossing $100B in assets in June 2021.

    - Average director age and background skews toward VC, tech and consulting rather than deep bank risk management. *(Exhibit 16)*
    - 93.9% of SVB's deposits were uninsured. This was far more than other banks like JPMorgan (52.5%) or B of A (14.1%). *(Exhibit 14)*
- **What asset-liability issues arose at SVB?**
    - On a book basis, assets were $212B against $213B liabilities+equity, with $13B in equity. *(Exhibit 13)*
    - On a fair-value basis, assets drop to $197B while deposits stay fixed at $173B, flipping equity to -$5B. *(Exhibit 13)*
    - SVB funded 82.9% of its balance sheet with deposits vs. 64.5–80.4% at the five largest US banks, with almost no long-term debt or diversified borrowing to cushion outflows. *(Exhibit 10)*

Q4.

- **Failure of Board Oversight and Accountability:** SVB's board of directors failed to oversee senior leadership and hold executives accountable for basic interest rate and liquidity risk management.
- **Absence of a Chief Risk Officer (CRO):** SVB operated without a CRO for a critical nine-month window between Apr 2022 and Jan 2023. This vacancy left the bank without executive-level risk oversight precisely when the Fed was accelerating its historic rate hikes and unrealized losses were accumulating.
- **Ignoring Internal Risk Metrics and Warnings:** SVB's leadership repeatedly failed its own internal corporate stress testing. At the time of insolvency, SVB had accumulated 31 open supervisory findings from regulators covering core governance flaws, liquidity vulnerabilities, and unhedged interest-rate exposure.
- **Inadequate Scaling During Rapid Growth:** When SVB crossed the $100 billion asset threshold in 2021, the board failed to proactively scale its risk-management infrastructure and compliance frameworks.

Q5.

- **New Risks That Have Emerged**
    - **Social-Media-Fueled Bank Runs:** Coordinated communication channels and virality on platforms like Twitter allowed panic to spread across an entire ecosystem instantaneously, turning localized anxiety into systemic capital flight in a matter of hours.
    - **Power of Digital Banking Liquidity Velocity:** With digital banking apps and instant online transfers, depositors can move massive amounts of money with a single tap. Historically, a severe bank run rarely exceeded 10% of total

deposits over multiple weeks. However, SVB lost 25% of its total deposits in just 10 hours.
    - **Socio-Economic Concentration in Client Base:** SVB's depositor base was tightly clustered in one interconnected niche: tech founders and VCs. When a few prominent VC heads issued warnings, the entire deposit base acted as a single herd.
- **The Impact for Financial Institutions**
    - **Adoption of 'Reverse Stress Testing':** Instead of assessing how a hypothetical recession would impact the balance sheet, institutions must now work backward by asking "what specific combination of deposit outflows and rate shocks would break the bank?"
    - **Recalibrating Liquidity Metrics:** Financial institutions can no longer treat uninsured corporate deposits as stable, long-term funding sources. Metrics like Liquidity Coverage Ratio (LCR) need to be re-evaluated.
    - **Tighter Regulatory Baselines for Regional Banks:** Lawmakers and central banks have pushed for strict regulatory supervision, rigorous standard stress testing, and higher capital buffers to apply uniformly across a much wider tier of institutions.

Q6.

- **Implement a Comprehensive Interest Rate Hedging Program**
    - SVB should have utilized interest rate swaps and derivatives to hedge its massive portfolio of fixed-rate long-term securities.
    - This recommendation would have eliminated the massive $1.8 billion after-tax loss when they were forced to liquidate securities.
- **Match Asset Duration to Liability Liquidity**
    - Instead of chasing smaller yield premiums by locking liquidity into long-term bonds, SVB should have maintained a short-duration asset strategy. They should have invested excess deposits in short-term Treasury Bills or floating-rate notes.
    - Short-term and floating-rate assets mature quickly and automatically adjust to higher yields when interest rates rise. This would have kept the asset portfolio highly liquid and protected its fair-market value from falling.
- **Establish Continuous Risk Leadership and Enforce Internal Risk Limits**
    - The board should have immediately filled the Chief Risk Officer (CRO) position and enforced the bank's internal interest rate risk limits and stress-testing mandates.
    - An active CRO and a risk committee would have acted upon the repeated failures of internal interest rate risk stress tests, and would have forced the bank to stop further long-term bond purchases.

- Addressing the 31 open supervisory findings would have improved core governance and liquidity vulnerabilities before market conditions deteriorated.

Q7.

- **Analysis of the Fed's Reaction**
  - Approving the systemic risk exception to guarantee 100% of uninsured deposits at SVB was a critical step. Otherwise, the situation would have triggered massive bank runs at other regional financial institutions.
  - Creating the BTFP allowed eligible banks to secure one-year loans by pledging high-quality assets at par value instead of market value. This addressed the liquidity lock that broke SVB.
  - Overall, I think the Fed's reaction was necessary and tactical. However, it also was mostly proactive instead of being reactive.
- **Key Issues and Assumptions to Re-Examine**
  1. **Inflation/Rate Shocks Occur in Isolation**
     - The Fed's historical stress-test scenarios did not adequately simulate an environment where interest rates rose as rapidly as they did simultaneously with a severe liquidity drain in a concentrated sector.
     - Stress testing parameters must be frequently updated to model highly correlated macroeconomic shocks.
  2. **Traditional Liquidity and Run Timelines Still Exist**
     - Supervisory frameworks long assumed bank runs take days or weeks to unfold, governed by historical deposit outflow speeds of 10% or less per day.
     - The Fed must adjust its liquidity requirements to acknowledge that modern digital banking apps and real-time social media communication have massively reduced the duration of bank runs to just a few hours. This demands the integration of "reverse stress testing" into standard oversight.
  3. **Tailoring Frameworks Are Sufficient for Regional Banks**
     - Following the 2018 regulatory rollbacks, the Fed assumed banks with assets between $100 billion and $250 billion posed less systemic risk and could safely be granted extended transition periods for compliance.
     - The Fed must re-evaluate its tailoring policy, recognizing that a regional bank's failure can carry massive systemic consequences irrespective of asset size if its business model or depositor base is highly concentrated.
